

Rajlaxmi Patil

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Professional Summary

Electronics and Telecommunication Engineering student with hands-on experience in embedded systems, IoT, and intelligent hardware solutions. Experienced in developing sensor-based projects, real-time monitoring systems, and automation applications. Eager to apply technical knowledge and engineering skills in embedded, automation, and industrial domains.

Education

AISSMS Institute of Information Technology, Pune	2023–2027
B.Tech in Electronics and Telecommunication Engineering	
CGPA: 8.05	
Higher Secondary Certificate (HSC)	72%
Secondary School Certificate (SSC)	98.7%

Technical Skills

Languages: C, Embedded C, Python (Basic).

Microcontrollers Embedded Systems: PIC18F4550, ESP32, Arduino, ARM Basics

Sensors Modules: HC-SR04 Ultrasonic Sensor, ADXL345 Accelerometer, GPS NEO-6M, MPU6050, GSM Module

Automation Control: PLC Fundamentals, SCADA Basics, Industrial Automation Concepts, Sensors and Actuators

Software Tools: Proteus, Arduino IDE, MATLAB (Basic), VS Code, Git, Multisim

IoT Applications: IoT Systems, Real-Time Monitoring, Sensor Interfacing, Wireless Communication.

Projects

1. Obstacle Avoiding Robot PIC18F4550, HC-SR04, Motor Driver

- Designed and developed an autonomous obstacle-avoiding robot using the PIC18F4550 microcontroller.
- Integrated ultrasonic sensors for obstacle detection and distance measurement.
- Implemented motor control algorithms for autonomous navigation.
- Interfaced LCD display and motor driver circuits for system monitoring and control.
- Technologies: PIC18F4550, HC-SR04, TB6612FNG, Embedded C.

2. Smart Emergency Response Grid

- Developed an IoT-based accident detection system using ESP32, GPS, and accelerometer sensors.
- Detected vehicle collisions using acceleration threshold analysis.
- Transmitted real-time location information and emergency alerts to emergency contacts.
- Developed interactive dashboards.

- Designed a monitoring dashboard for emergency response and traffic management.
- Implemented real-time sensor data acquisition and wireless communication. : ESP32, ADXL345, GPS NEO-6M, IoT, Firebase

3. Neuromorphic Motion Prediction and Surveillance System

- Designed and developed a real-time neuromorphic-inspired motion prediction system using ESP32-CAM and Python.
- Implemented motion detection using frame differencing and contour analysis techniques.
- Developed object tracking, speed estimation, and future position prediction algorithms for moving objects.
- Integrated buzzer alerts and real-time event logging for motion detection and monitoring applications.
- Technologies: ESP32-CAM, Python, OpenCV, NumPy, Computer Vision, WiFi Communication

Achievements

Runner-Up – Techathon 2025 National Level Hackathon

Built multiple AI Integrated and IOT integrated electronics projects.

Interests

- Embedded Systems
 - Internet of Things
 - Industrial Automation
 - Artificial Intelligence Applications
 - Sensor Interfacing and Real-Time Systems
 - Intelligent Healthcare Technologies
 - Robotics and Automation